

Silent Cart: FreshBasket Loyalty Churn Prediction & Retention Strategy

Final report · Data Science · Prepared with the reproducible pipeline in this repository (`python run_pipeline.py`)

Executive summary

- **Most of the churn being reported has already happened.** Of the 2,368 churn-eligible members, 34.3% are labelled churned. But **80% of those "churners" had already stopped buying before April:** they had made no purchase for 3+ months. They need a *win-back* programme, not a prediction model.
- **The real prediction problem is among Active members** (bought in Jan-Mar 2024): 1,715 members, of whom 9.6% churned. Here a recency rule gets a PR-AUC of only 0.46; our **logistic regression** reaches **0.79** on an untouched, out-of-time test quarter. At the chosen threshold it catches **76% of Active churners at 70% precision** (base rate 9.6%). The top 10% of scores hold 7.5x the average churn rate.
- **Churn has a visible fingerprint.** In the ~2 months before their last purchase, churners' transactions, app sessions and email opens fall sharply while support tickets spike. Members with 3+ tickets in 6 months churn at 37%, against 4% for those with none. Members whose spend fell 40%+ quarter-on-quarter churn at 37%.
- **Tier does not protect.** Platinum members churn as often as Silver (34% vs 35%). Age, tenure and tier are not statistically related to churn among Active members. Behaviour, engagement and support experience are.
- **Targeting beats blanket offers.** Today's approach, the same coupon for everyone, costs \$23,680 for this cohort and loses \$15,423 under our stated assumptions. A **next-best-action playbook** (personalised coupon, proactive service call or win-back, chosen per member by expected value) costs \$3,860 (84% less), prevents about 43 churners (73% of the blanket offer's saves) and returns **\$3,559**. That is a swing of about \$80,161 per 10,000 eligible members per quarter.

Recommendation: split the churn KPI into *Lapsed* (win-back) and *At-risk Active* (retention). Score members monthly, route each one to its next best action, and prove the uplift with a 20% hold-out control group before scaling.

1. Business question and approach

FreshBasket sends the same retention offers to every member, which wastes money on members who were never going to leave. Management asked four questions: **who** will churn, **which signals** come first, whether **tier, marketing engagement and support experience** matter, and **what action** would reduce churn.

The pipeline runs in eight steps: **data quality** → **leakage-safe features** → **rolling-origin validation** → **four models compared** → **calibration and threshold tuning** → **SHAP drivers and segments** → **ablation** → **retention economics**. It is a reproducible Python package (`src/`) with one entry point (`run_pipeline.py`),

MLflow experiment tracking, unit tests (including an automated leakage test), a FastAPI scoring endpoint and a PySpark port of the feature pipeline for Databricks.

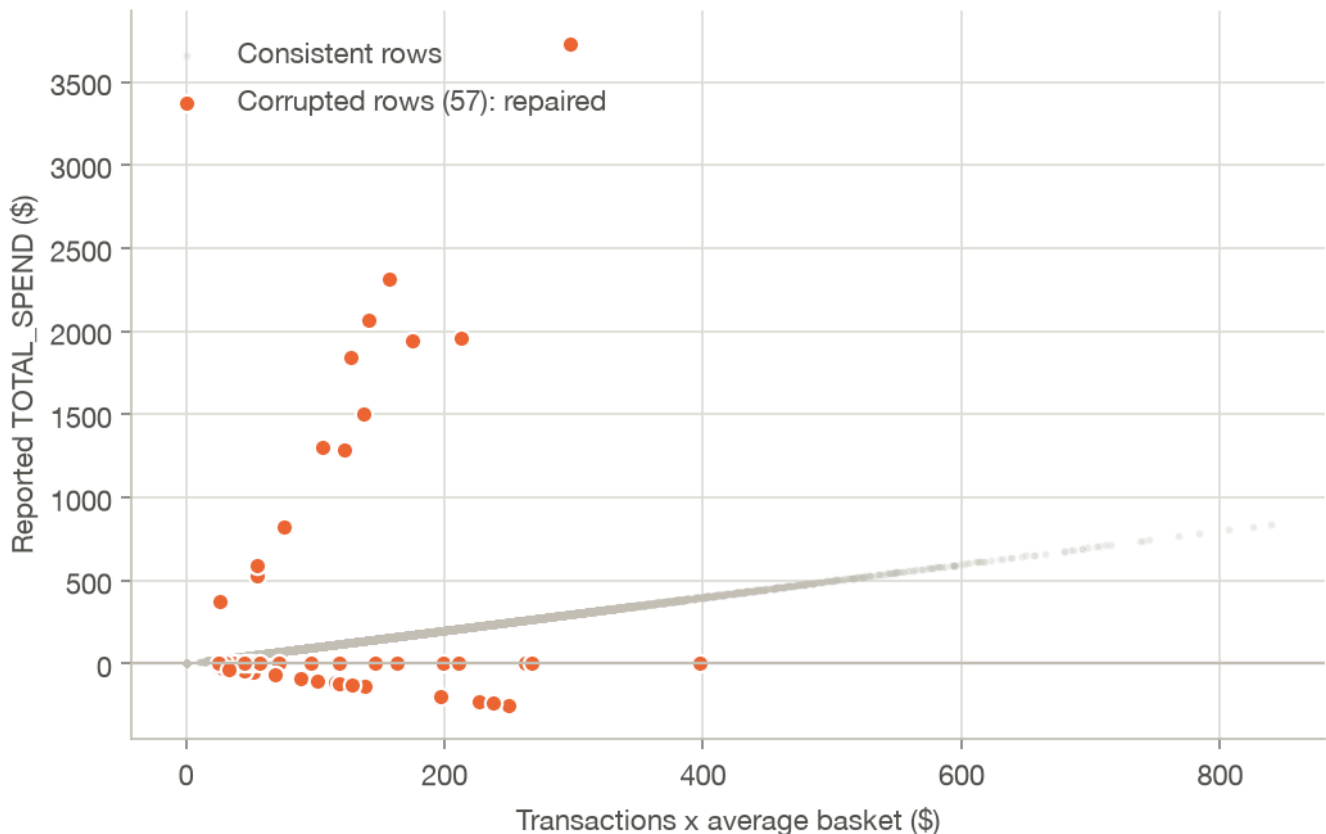
2. Data quality: what was wrong and how it was handled

The workbook has three tables: 2,600 members, 28k member-months, Jan-2023 to Jun-2024. Every sheet carries a title banner, so the real header is on row 2. Reading with defaults silently yields `Unnamed` columns. All 28 checks are logged in `outputs/data_quality_log.csv`. The issues found:

Table	Issue	Rows	Handling
profile	Exact duplicate rows	8	Dropped
profile	CITY: inconsistent casing / whitespace	104	Stripped and title-cased (40 raw -> 10 clean values)
profile	MEMBERSHIP_TIER: inconsistent casing / whitespace	104	Stripped and title-cased (12 raw -> 3 clean values)
profile	AGE missing	78	Kept as missing; median-imputed at feature time + AGE_MISSING flag
profile	GENDER missing	52	Filled with explicit 'Unknown' category
profile	HOME_STORE_PRICE_TIER missing	130	Filled with explicit 'Unknown' category
activity	Exact duplicate rows	58	Dropped
activity	Conflicting duplicates (same customer-month, spend sign flipped)	4	Kept the row with positive spend
activity	Negative TOTAL_SPEND with transactions > 0	22	Rebuilt as TRANSACTIONS x AVG_BASKET_VALUE
activity	Zero TOTAL_SPEND with transactions > 0	20	Rebuilt as TRANSACTIONS x AVG_BASKET_VALUE
activity	Outlier TOTAL_SPEND (> 3x transactions x basket)	13	Rebuilt as TRANSACTIONS x AVG_BASKET_VALUE (max was \$3,731)
activity	Remaining high spend after repair (> Q3 + 3 IQR)	60	Kept: consistent with transactions x basket, so genuine big shoppers
activity	Zero-transaction months (engagement but no purchase)	2,330	Kept: valid inactivity signal; spend/basket/promo already 0
activity	Zero-transaction months reporting DISTINCT_CATEGORIES > 0	1,452	Set DISTINCT_CATEGORIES to 0 (cannot buy categories without a transaction)
activity	Zero-transaction months reporting COUPONS_REDEEMED > 0	653	Set COUPONS_REDEEMED to 0 (a redemption requires a transaction)
activity	Missing months inside a member's active span (769 members)	953	Not imputed as zero: treated as missing data; MISSING_MONTHS_6M feature added
activity	Profiled members with no activity rows at all	6	Cannot be scored; excluded from modelling
label	Exact duplicate rows	5	Dropped
label	LAST_PURCHASE_DATE missing (member never purchased)	120	Not churn-eligible by definition; excluded from modelling
label	CHURNED disagrees with label re-derived from activity (99.7% agree)	9	Official label kept; mismatches trace to missing activity rows
label	CHURNED=1 but no pre-April purchase visible in activity	5	Excluded (no history to build features from)
label	Leaky columns: LAST_PURCHASE_DATE, MONTHS_OBSERVED, INSUFFICIENT_HISTORY_FLAG	3	Computed with Apr-Jun 2024 data; never used as features. Rebuilt from pre-cutoff data instead

Spend errors are detectable because $\text{spend} = \text{transactions} \times \text{basket}$

99.8% of rows satisfy the identity; the rest are negative, zero or 9-15x inflated and were rebuilt



Leakage audit. Three label-table columns, `LAST_PURCHASE_DATE`, `MONTHS_OBSERVED` and `INSUFFICIENT_HISTORY_FLAG`, are calculated over the full window *including* Apr-Jun 2024. Churners stop generating rows, so these columns partly encode the answer. They are never used. Leakage-safe equivalents (`recency_months`, `history_months`, `short_history`) are rebuilt from pre-cut-off data. Section 5 shows that including the leaky columns would inflate Active-member PR-AUC from 0.80 to 0.87.

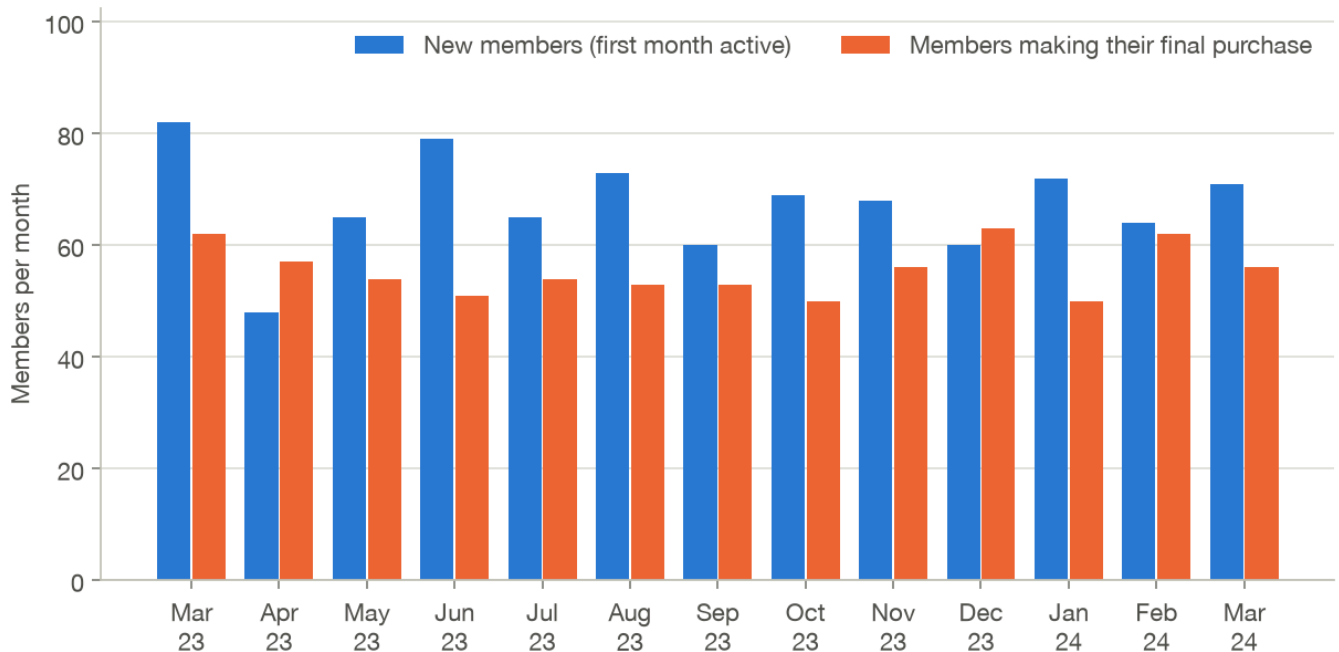
3. What churn really looks like

3.1 A leaky bucket hidden by sign-ups

Every month roughly 55 members make their final purchase, while new sign-ups keep headline member counts rising. The loss is invisible in the totals.

A leaky bucket: ~55 members make their final purchase every month

Mar-23 to Mar-24: 721 members bought for the last time vs 876 new arrivals; sign-ups hide the leak in headline counts

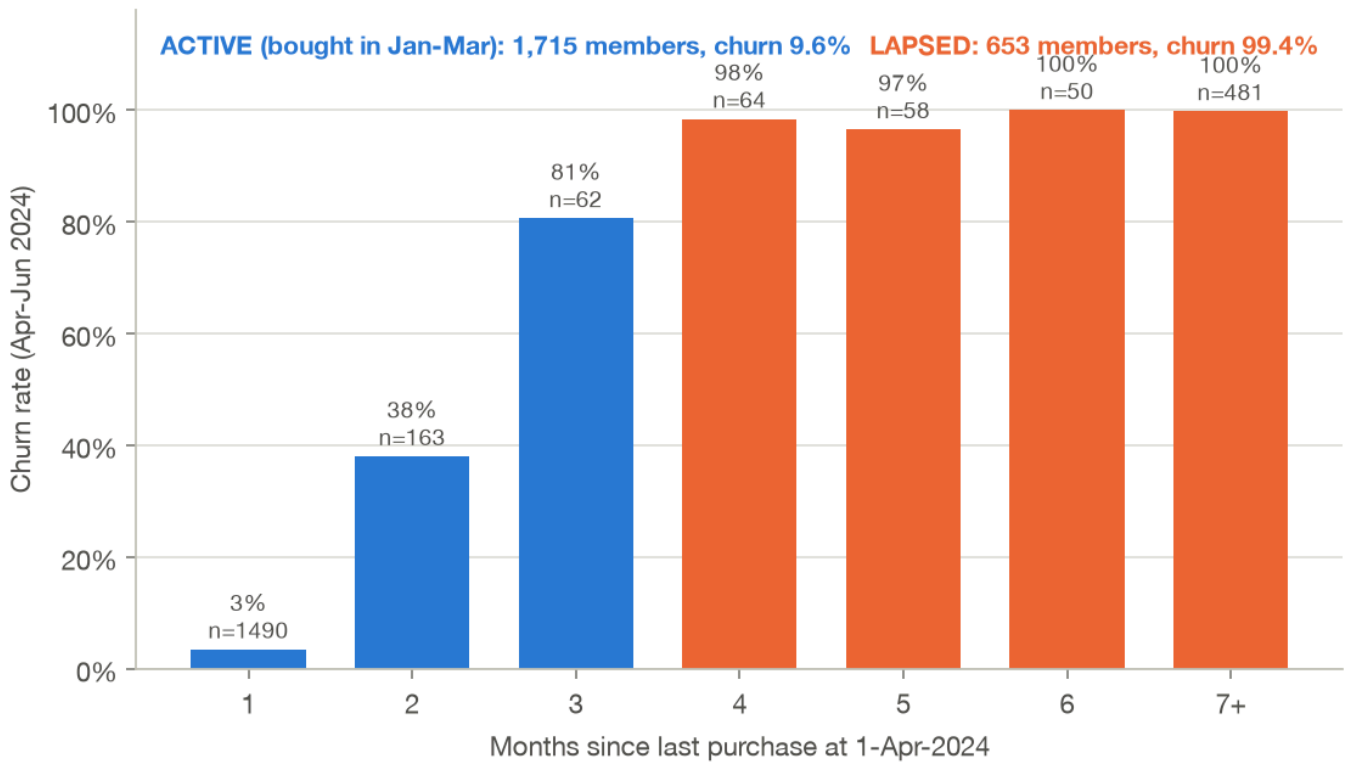


3.2 Two churn problems inside one label

Churn probability is a cliff in recency. Members who last bought 1 month before April churn at 3%; 2 months, 38%; 3 months, 81%; 4+ months, about 100%. The label combines members who left months ago with members who are about to leave.

Two different churn problems hide inside one label

80% of 'churners' had already stopped buying before April; only the Active group needs prediction



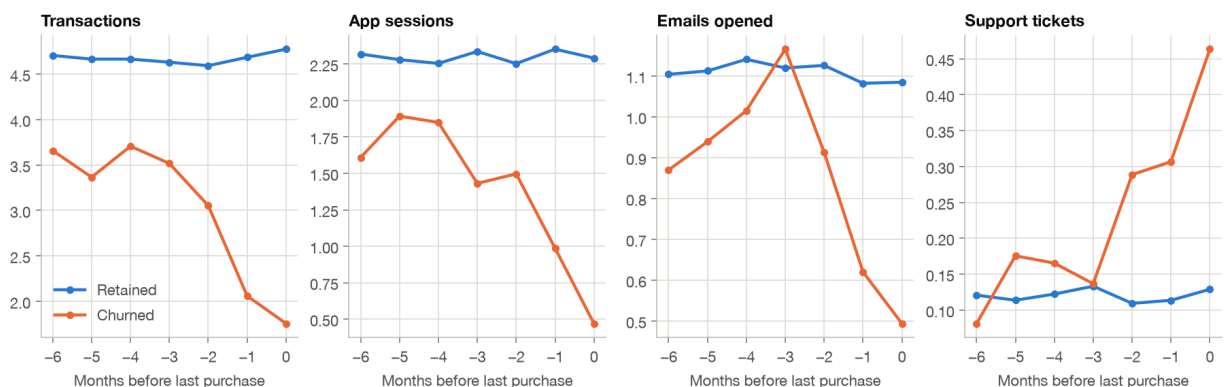
Segment	Members	Churn rate	Share of churners	What it needs
Lapsed (no purchase Jan-Mar 2024)	653	99.4%	80%	Win-back campaign; a rule identifies them
Active (purchased Jan-Mar 2024)	1,715	9.6%	20%	Early warning model + retention action

3.3 The churn fingerprint

Lining up Active members on their last pre-April purchase shows *how* churn unfolds. Purchases, app sessions and email opens fade over the final ~2 months while support tickets spike. Those ~2 months are the intervention window.

Churn fingerprint: purchases, app use and email opens fade while support tickets spike

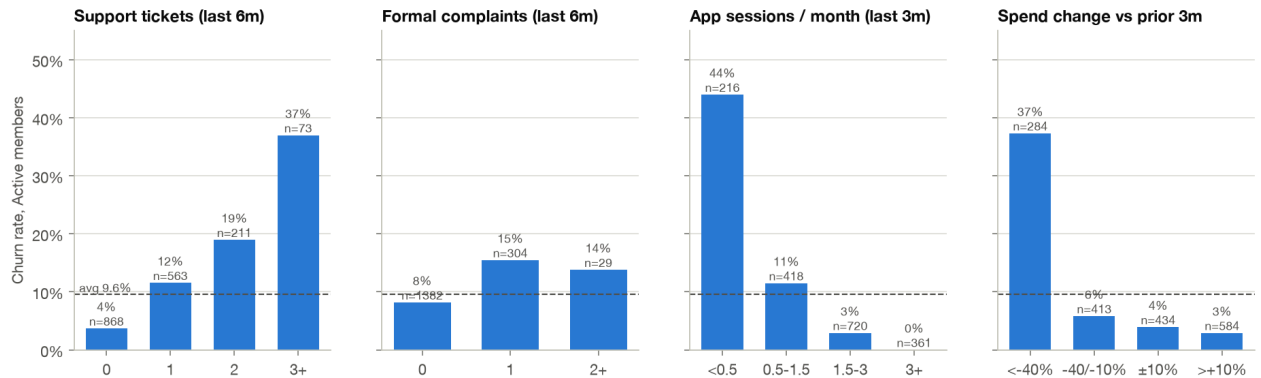
Active members, monthly averages aligned on each member's last purchase before April 2024 (t = 0). The decline starts ~2 months before the last purchase; that is the intervention window



4. Engagement and support experience vs churn

Support friction and fading engagement both raise churn risk among Active members

Observed churn rate (Apr-Jun 2024) by pre-April signal; dashed line = Active-segment average



Statistical evidence for Active members: Mann-Whitney U tests, Benjamini-Hochberg corrected. Rank-biserial $r > 0$ means churners score higher.

Group	Feature	Mean (churned)	Mean (retained)	Effect size (r)	q (BH)
Behavioural	txn_l3	1.35	4.60	-0.84	<0.001
Engagement	app_l3	0.58	2.27	-0.75	<0.001
Behavioural	categories_l3	1.11	2.57	-0.66	<0.001
Behavioural	txn_change_pct	-0.41	0.11	-0.63	<0.001
Behavioural	recency_months	1.99	1.08	+0.63	<0.001
Behavioural	spend_change_pct	-0.36	0.12	-0.62	<0.001
Behavioural	active_months_l3	1.77	2.66	-0.59	<0.001
Engagement	coupons_l3	0.18	0.59	-0.55	<0.001
Engagement	app_change_pct	-0.40	0.16	-0.51	<0.001
Engagement	email_l3	0.43	1.09	-0.49	<0.001
Support	tickets_6m	1.49	0.63	+0.43	<0.001
Support	tickets_l3	1.02	0.33	+0.40	<0.001

- **Engagement:** churners opened 0.4 emails/month vs 1.1, and used the app 0.6 vs 2.3 sessions/month ($r = -0.75$). Members with fewer than 0.5 app sessions/month churn at 46%.
- **Support:** churners raised 1.5 tickets in 6 months vs 0.6. A formal complaint in the last quarter roughly doubles churn (17% vs 9%; chi-square $p = 0.0004$).
- **Not significant:** tier, age and tenure ($q > 0.6$). Marketing opt-in is borderline ($q = 0.06$).

5. Modelling and validation

Validation design (no random split). The churn definition can be replayed at any cut-off, so we built forward-chaining snapshots:

Role	Cut-off	Label window	Members	Churn	Active churn
Train	Jul-2023, Oct-2023	Jul-Sep, Oct-Dec 2023	3,767 rows	21.5%	10.2%
Validation (model choice, calibration, threshold)	Jan-2024	Jan-Mar 2024	2,167	30.1%	10.2%
Test (reported once)	Apr-2024	Apr-Jun 2024, official label	2,368	34.3%	9.6%

Models compared: - (a) a **recency rule** (rule-based baseline: churn rate by months since last purchase); - (b) **logistic regression** with class weights; - (c) **random forest** with balanced class weights; - (d) **gradient boosting** with balanced sample weights.

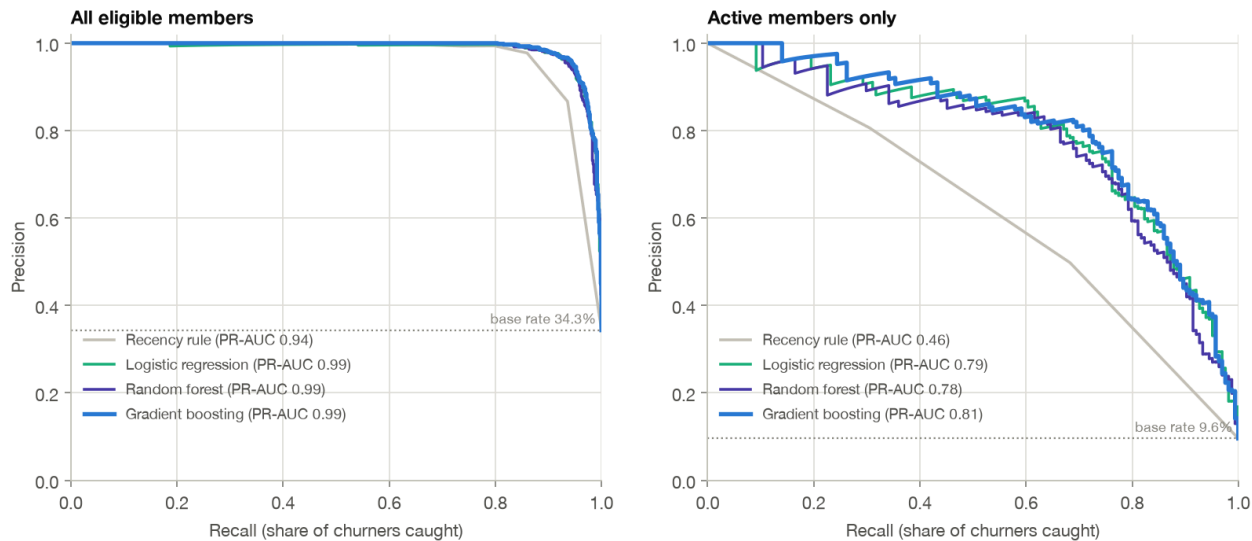
Hyper-parameters were chosen on validation **Active-member PR-AUC**, since that is where prediction is hard. Re-weighted scores were then Platt-calibrated on validation, so outputs are real probabilities. The threshold maximises validation Active-member F1.

Results on the untouched Apr-Jun 2024 test quarter:

Model	Population	Precision	Recall	F1	PR-AUC (95% CI)	ROC-AUC	Accuracy	Lift @ top 10%
Recency rule	All eligible	0.867	0.936	0.900	0.944 (0.93-0.96)	0.961	0.929	2.9x
Recency rule	Active only	0.498	0.683	0.576	0.464 (0.39-0.54)	0.814	0.904	5.7x
Logistic regression	All eligible	0.930	0.952	0.941	0.986 (0.98-0.99)	0.991	0.959	2.9x
Logistic regression	Active only	0.698	0.762	0.729	0.794 (0.73-0.85)	0.961	0.946	7.5x
Random forest	All eligible	0.918	0.956	0.937	0.987 (0.98-0.99)	0.991	0.956	2.9x
Random forest	Active only	0.663	0.780	0.717	0.775 (0.71-0.83)	0.955	0.941	7.4x
Gradient boosting	All eligible	0.934	0.953	0.943	0.989 (0.98-0.99)	0.992	0.961	2.9x
Gradient boosting	Active only	0.708	0.768	0.737	0.807 (0.75-0.86)	0.962	0.948	7.6x

Precision-recall on the untouched Apr-Jun 2024 test cohort

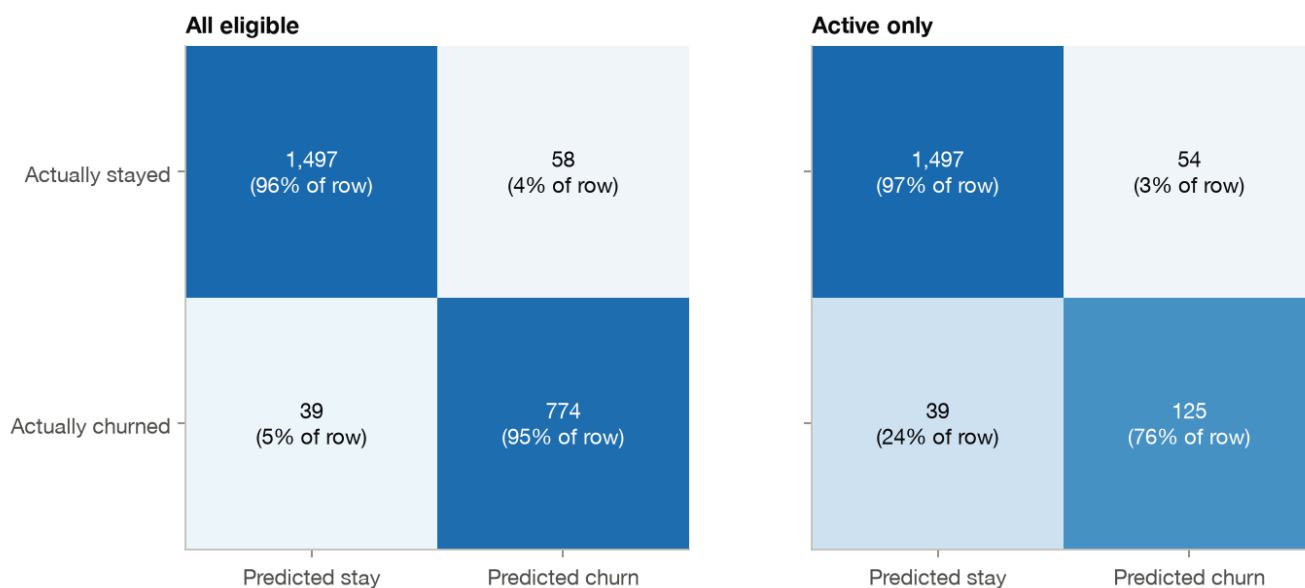
On all members every model looks strong (lapsed members are easy); the Active panel is the real test, where ML clearly beats the recency rule



- On *all* members every model scores above 0.94 PR-AUC, because lapsed members are trivially predictable. **The Active panel is the real test:** machine-learning models roughly double the recency rule's PR-AUC (0.46 to about 0.8).
- Logistic regression was selected on validation.** Gradient boosting scores slightly higher on test (0.807 vs 0.794), but the bootstrap intervals overlap almost entirely: the models are statistically tied. The transparent, cheaper model is deployed, and gradient boosting is kept as a challenger and SHAP cross-check.

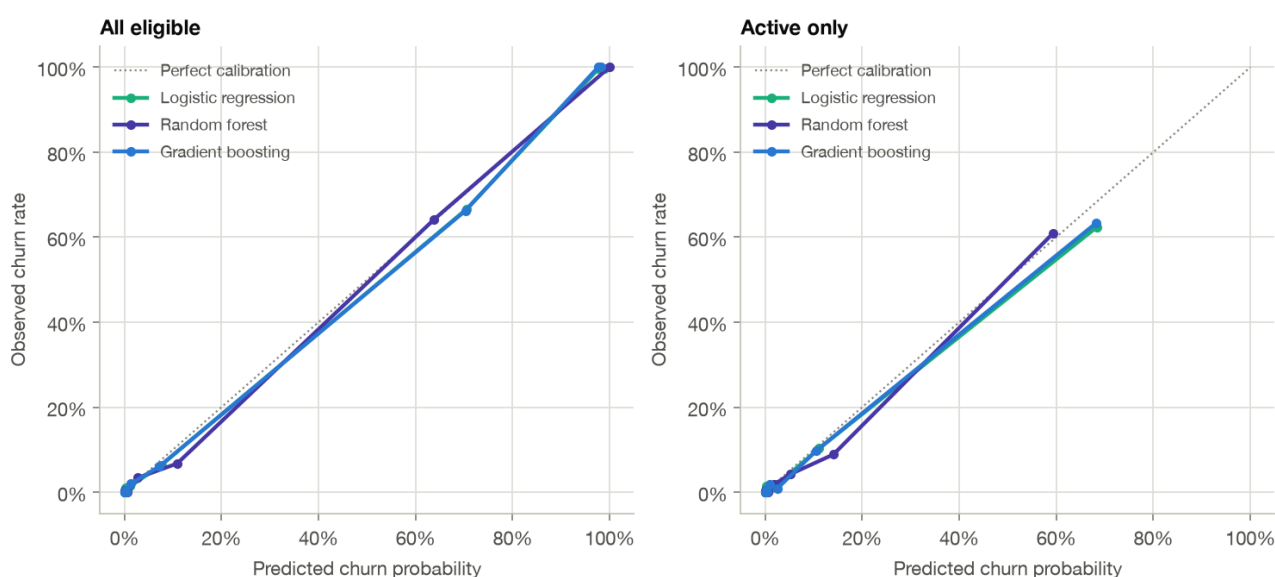
Confusion matrix: Logistic regression at threshold 0.37

Threshold maximises F1 for Active members of the Jan-2024 validation snapshot; applied unchanged to the test cohort



Calibration on the test cohort: probabilities can be used as real churn odds

Platt scaling fitted on the validation snapshot undoes the class re-weighting, which matters for value-at-risk maths



Validation-design checks (5-fold random CV on the test cohort, shown only for comparison):

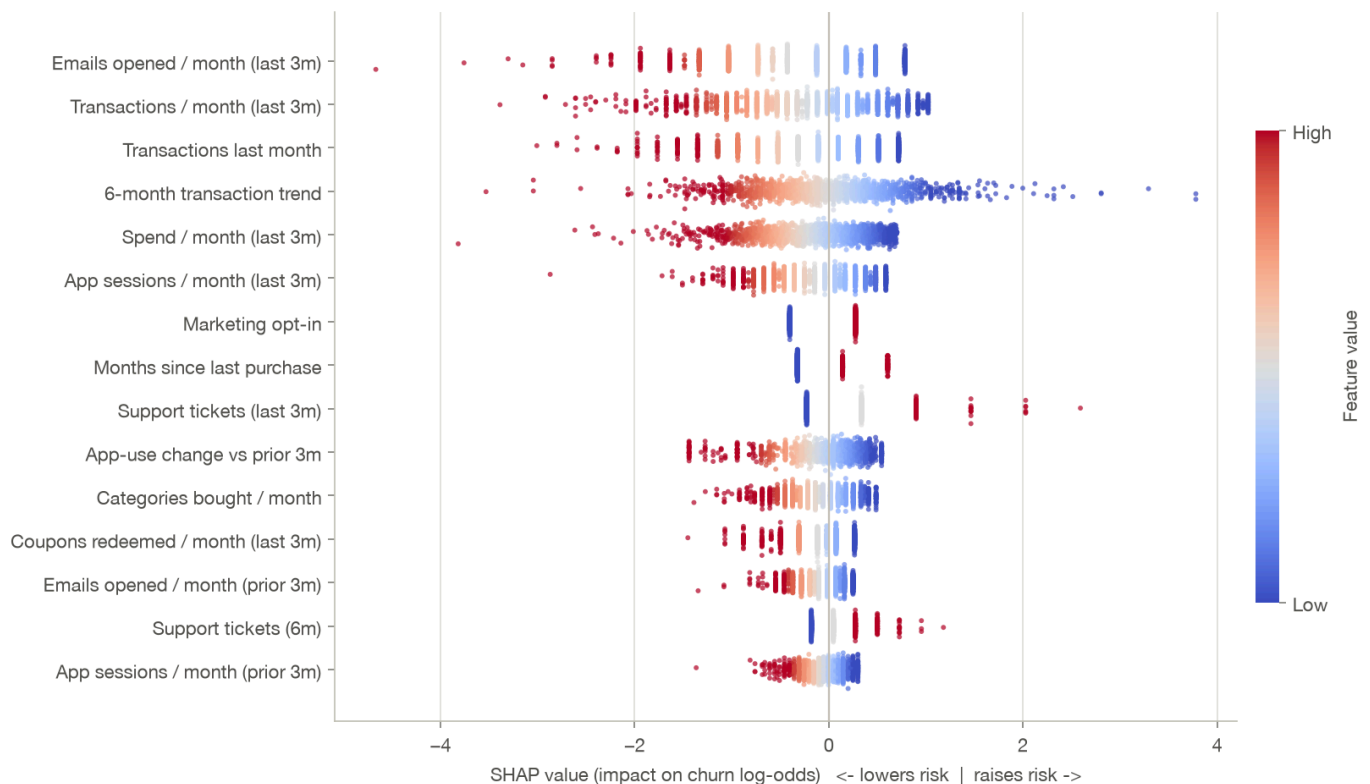
Set-up	PR-AUC all	PR-AUC Active
Out-of-time test (reported), clean features	0.989	0.807
Random 5-fold CV, clean features	0.987	0.800
Random 5-fold CV, + leaky label columns	0.992	0.872

Random CV scores close to the out-of-time result on this dataset, so behaviour is stable over time. The leaky label columns, however, would have inflated Active-member PR-AUC by about 0.07.

6. Churn drivers (SHAP) and segment risk

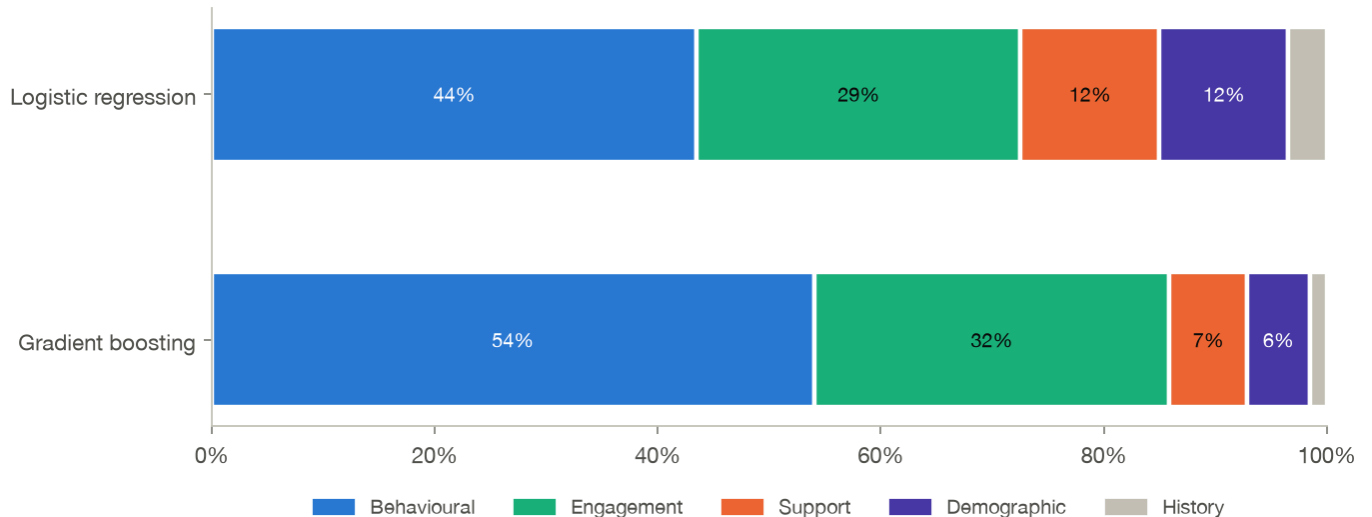
What drives churn risk for Active members (Logistic regression)

Low recent activity and engagement (blue dots on the right) and more support tickets push risk up



Where the predictive signal comes from (Active members)

Share of mean |SHAP| by feature group; both models agree that behaviour, engagement and support dominate



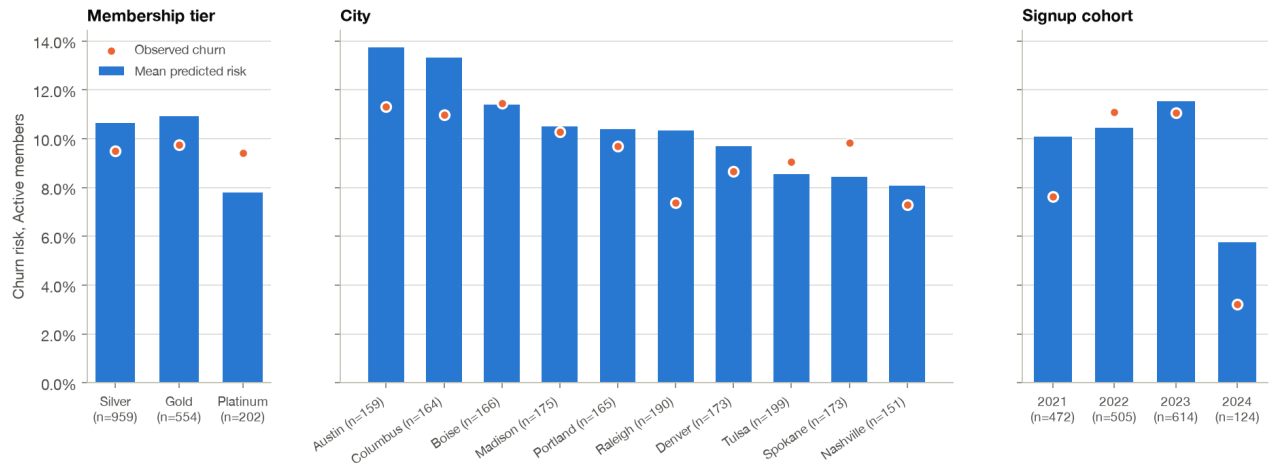
The strongest drivers for Active members are emails opened / month (last 3m), transactions / month (last 3m), transactions last month, 6-month transaction trend, spend / month (last 3m), app sessions / month (last 3m). Behaviour and engagement account for 73% of attribution in the logistic regression and 86% in gradient boosting. Support is the next layer; demographics add little. In the logistic model, *marketing opt-in* raises risk once email opens are accounted for: members who opted in but stopped opening emails are disengaging. This is a conditional effect, not a reason to avoid opt-ins.

Every member gets three plain-English, risk-increasing drivers in `outputs/churn_predictions.csv`. Examples from the high-risk Active list:

Member	Churn probability	Top 3 drivers (SHAP)
502373	100%	4 support tickets in last 3m; Transactions trending down (-1.8/month); 1.0 transactions/month (last 3m)
501335	100%	3 support tickets in last 3m; Transactions trending down (-1.5/month); 0.5 transactions/month (last 3m)
501338	99%	3 support tickets in last 3m; Transactions trending down (-1.3/month); 0.3 transactions/month (last 3m)
500989	99%	4 support tickets in last 3m; 1.0 transactions/month (last 3m); Transactions trending down (-0.9/month)
501056	99%	0.7 transactions/month (last 3m); No marketing emails opened in last 3m; Transactions trending down (-0.9/month)

Segment risk among Active members: differences are modest; behaviour matters more than who you are

Bars = mean calibrated churn probability; dots = observed Apr-Jun 2024 churn. Use segments for targeting logistics, not as risk proxies



Segments. Among Active members, predicted risk ranges from 8% (Nashville) to 14% (Austin). Austin and Columbus are the highest-risk cities. Platinum members score slightly lower predicted risk, yet *observed* churn is flat across tiers. Members who joined in 2024 churn least (honeymoon period). Segment differences are much smaller than behavioural ones, so segments should guide *logistics* (which store team, which channel), not who is targeted.

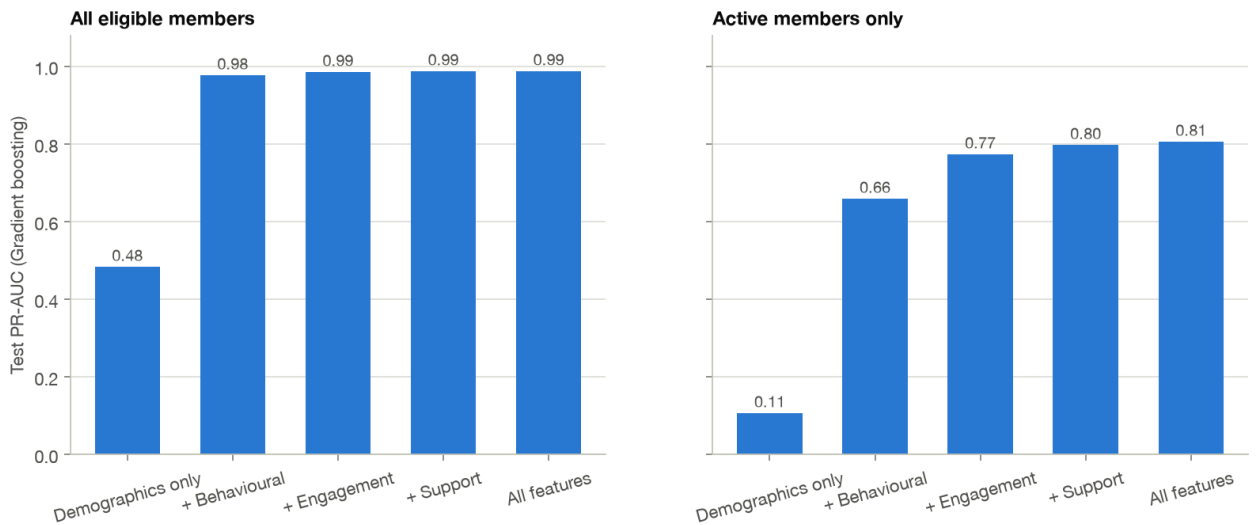
7. Ablation: do behaviour, engagement and support add value over demographics?

Each step adds a feature group cumulatively (gradient boosting, test quarter):

Uplift to detect	Control members	Flagged members needed	Quarters at this volume	Eligible base for 1 quarter
10%	324	1,619	9.0	21,413
15%	146	729	4.1	9,642
20%	83	415	2.3	5,478
25%	54	267	1.5	3,530
30%	38	186	1.0	2,461

Ablation: behaviour carries the signal; demographics alone are close to random

Active-member PR-AUC rises 7.5x from demographics-only to all features; see ablation_study.csv for the no-recency variant



- **Demographics alone are no better than chance** for Active members (PR-AUC 0.11 vs a 9.6% base rate).
- **Behavioural** features carry most of the signal.
- **Engagement** adds +0.11 PR-AUC.
- **Support** adds +0.02.
- **Without recency** the final model still reaches 0.80, so the early-warning signal is not just "hasn't bought lately".

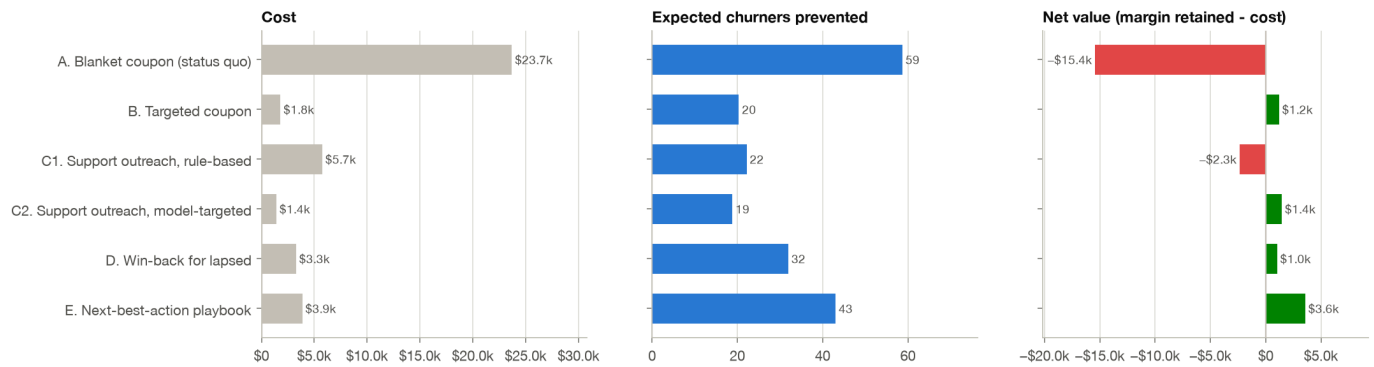
8. Retention scenarios

Expected churners prevented = churn probability x relative uplift. Net value = margin retained – cost. All assumptions are listed in Section 10, and the uplifts are deliberately varied in the sensitivity chart.

Scenario	Members	Churners prevented	Cost	Margin retained	Net value	ROI	Break-even uplift
A. Blanket coupon (status quo)	2368	58.7	\$23,680	\$8,257	–\$15,423	-65%	21%
B. Targeted coupon	179	20.4	\$1,790	\$2,983	\$1,193	+67%	9%
C1. Support outreach, rule-based	383	22.2	\$5,745	\$3,398	–\$2,347	-41%	42%
C2. Support outreach, model-targeted	94	18.8	\$1,410	\$2,858	\$1,448	+103%	12%
D. Win-back for lapsed	653	32.0	\$3,265	\$4,283	\$1,018	+31%	4%
E. Next-best-action playbook	549	43.0	\$3,860	\$7,419	\$3,559	+92%	n/a

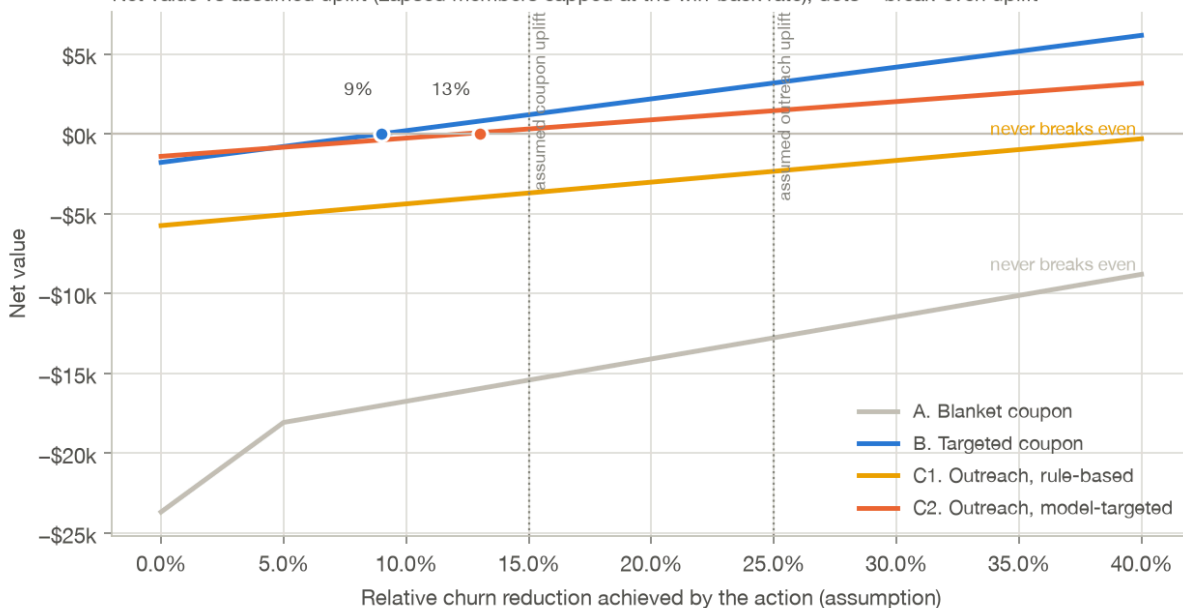
Next-best-action targeting: 84% less spend, 73% of the saves, and a profit instead of a loss

Assumptions: coupon \$10 at 15% relative uplift; service call \$15 at 25%; win-back \$5 at 5% reactivation; value = 6 months of spend x 25% margin



Sensitivity: targeted actions break even at about half the assumed effect; blanket never does

Net value vs assumed uplift (Lapsed members capped at the win-back rate); dots = break-even uplift



- **Blanket coupon (status quo)** never breaks even. Most of the money goes to members who were staying anyway, or who have already left.
- **Support outreach**, as the brief suggested, loses money if every member with tickets is called (–\$2,347). The same calls **restricted to model-flagged members** return \$1,448 (ROI +103%) and reach 85% of the saves with a quarter of the calls.
- **A model-based what-if supports this.** Re-scoring the 383 support-issue members with their tickets "resolved" lowers mean risk from 23% to 14% (41% relative). This is associational, so treat it as an upper bound; our planning assumption is 25%.
- **The targeted coupon** breaks even at a 9% uplift, against 21% for the blanket coupon.
- **Next-best-action playbook (recommended):** each member receives at most one action, and only when its expected value is positive:

Recommended action	Members	Expected net value	6-month revenue at risk
Monitor (no paid action)	1819	\$0	\$92,889
Personalised coupon	59	\$562	\$30,709
Proactive service call	82	\$1,527	\$44,106
Win-back offer	408	\$1,540	\$280,895

The playbook lowers expected churn among Active members from 10.4% to 9.1% next quarter and recovers about 32 lapsed members, for a net \$3,559 on this cohort. The full ranked list, with action, value at risk and drivers per member, is in `outputs/retention_action_list.csv`.

9. Beyond the brief: advanced analytics

9.1 Deep-learning challenger (champion / challenger)

A GRU recurrent network (PyTorch) reads each member's raw 6-month activity sequence instead of hand-built trends. Same splits, weighting, calibration and threshold logic. It reaches **0.825** Active PR-AUC (95% CI 0.76-0.87) vs 0.794 for the deployed model, with 80% precision and 78% recall. A **paired bootstrap** puts the gain at +0.031 (95% CI -0.001 to +0.067; GRU better in 97% of resamples): likely real, but not yet proven.

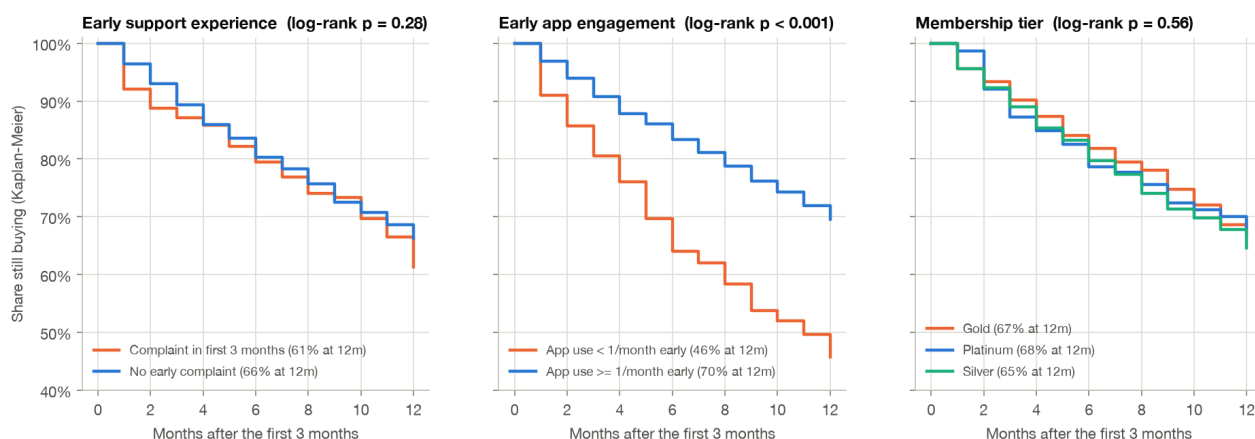
Decision: keep the transparent logistic model live (its per-member SHAP reasons drive the action list), shadow-score the GRU for one quarter, and promote it if the gain holds. This is a standard champion/challenger set-up.

9.2 When do members leave? Survival analysis

Kaplan-Meier curves of time to final purchase, landmarked at month 3 (groups are defined on each member's first 3 months, which avoids immortal-time bias), with log-rank tests: - **Early app use:** members using the app less than once a month early on are still buying at 12 months only 46% of the time, vs 70% ($p < 0.001$). App onboarding is a lifetime-value lever. - **No difference:** early complaints ($p = 0.28$) and tier ($p = 0.56$). Support friction predicts churn when it is *recent* (section 4), not early in the relationship.

Survival: low early app use cuts 12-month retention to 46% (vs 70%); early complaints ($p=0.28$) and tier ($p=0.56$) do not

Time to final purchase, landmarked at month 3. Support friction matters when it is RECENT (see churn model), not early in the relationship

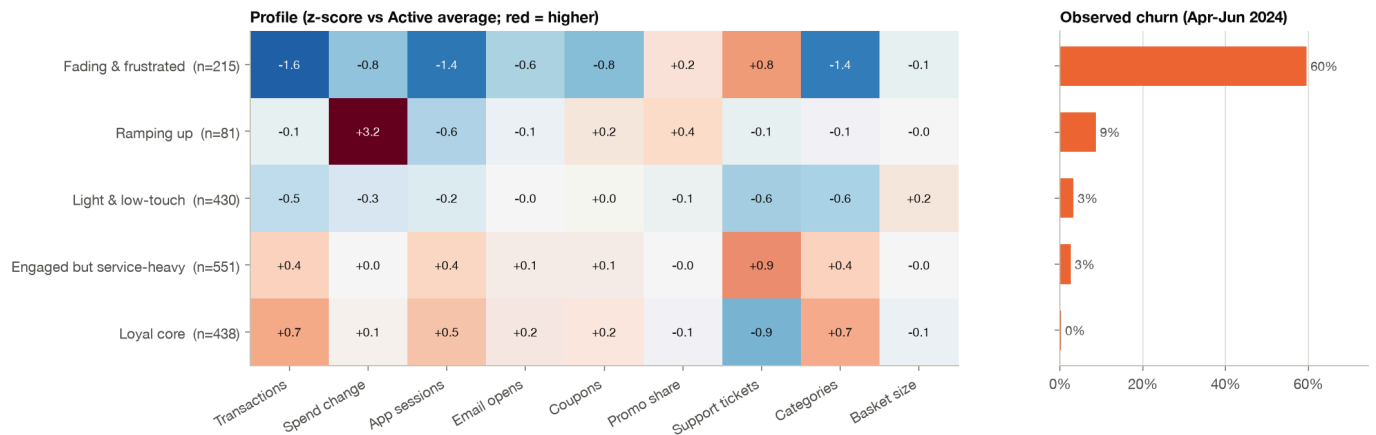


9.3 Behavioural personas

K-means on Active members' recent behaviour gives five messaging personas. Separation is modest (silhouette 0.12), so personas tailor the *message* while the churn score decides *who* gets it.

Persona	Members	Observed churn	Mean predicted	Recommended play
Fading & frustrated	215	60%	62%	Service-recovery call first, then a preferred-category win-back coupon
Ramping up	81	9%	10%	Reinforce the new habit: app onboarding and a 'next visit' reward
Light & low-touch	430	3%	6%	Cross-category offers to deepen the basket; low-cost digital nudges
Engaged but service-heavy	551	3%	2%	Fix friction fast (ticket follow-up); no discount needed while they still shop
Loyal core	438	0%	0%	No discounts; recognition and referral rewards

Five behavioural personas: 'Fading & frustrated' members churn at 60%
K-means on Active members' recent behaviour (silhouette 0.12: soft groups, used to tailor the message, not to replace the churn score)

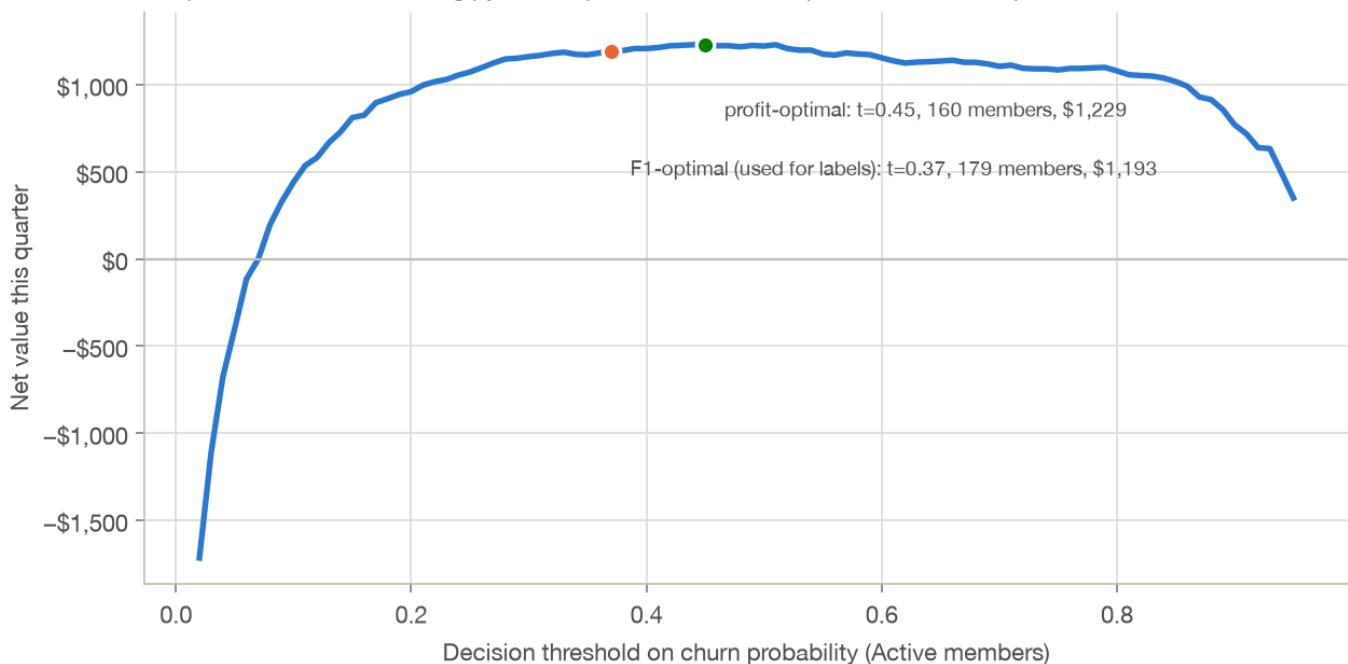


9.4 Profit-optimal threshold

Choosing the cut-off by money instead of F1 gives t = 0.45 (\$1,229) vs t = 0.37 for F1. The profit curve is flat across a wide band, so the targeting decision is robust to the exact threshold.

Profit is flat for thresholds 0.27-0.71: the targeting decision is robust

Coupon economics from config.py. Profit-optimal $t=0.45$ vs F1-optimal $t=0.37$: \$35 apart



9.5 Designing the experiment that replaces our assumptions

The uplift assumptions should be measured, not argued about. Hold out 20% of flagged members as a control group and use a two-sided two-proportion test ($\alpha = 0.05$, power 80%), baseline churn 76% among flagged members:

Uplift to detect	Control members	Flagged members needed	Quarters at this volume	Eligible base for 1 quarter
10%	324	1,619	9.0	21,413
15%	146	729	4.1	9,642
20%	83	415	2.3	5,478
25%	54	267	1.5	3,530
30%	38	186	1.0	2,461

Detecting the assumed 15% uplift needs about 729 flagged members (146 in control). At this sample's volume (2,368 eligible members) that takes about 4 quarters. A programme with roughly 9,642+ eligible members could read the result within a single quarter.

9.6 Next quarter: who is likely to churn in Jul-Sep 2024

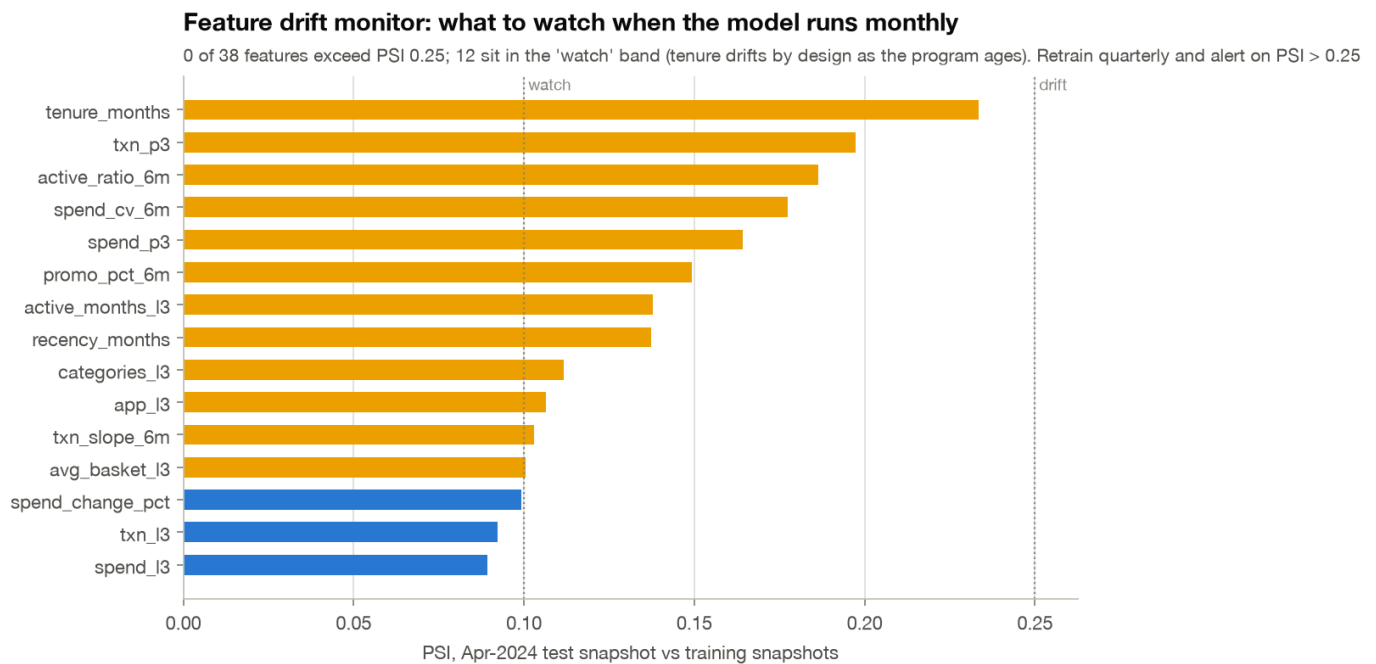
The model is used as it would be in production: a snapshot at Jul-2024 built from Jan-Jun 2024 behaviour, with no label yet.

Segment	Members	Expected churners (90% interval)	Flagged high-risk	6-month revenue at risk
Active	1,657	149 (137-161)	144	\$90,919
Lapsed	817	800	817	\$445,324

The ranked list with drivers and next best action is in `outputs/next_quarter_watchlist.csv`, and in the dashboard.

9.7 Monitoring: drift and fairness

- **Drift:** no feature exceeds PSI 0.25 (max 0.23, `tenure_months`, which drifts by design as the program ages). 12 features sit in the 0.1-0.25 watch band, supporting quarterly retraining.
- **Fairness:** recall and false-positive rates are similar across age bands and genders. Platinum recall is lower on few churners; it is recorded as a watch item in `MODEL_CARD.md`.



9.8 Silent Cart Retention Console (Dash)

`python app.py` opens an interactive console with four tabs: - **Overview:** headline KPIs. - **Members:** filter today's action list or next quarter's watchlist, then click any member to see their churn probability, SHAP reasons, persona, next best action and 18-month activity. - **Scenario simulator:** nine live assumption sliders, running the same scenario engine as this report. - **Insights:** survival, profit curve, drift, fairness and A/B sizing.

Dash is natively supported on Databricks Apps, so the console can be deployed next to the model.

10. Recommendations

1. **Redefine the churn KPI.** Report *Lapsed* (no purchase 3+ months: win-back) separately from *At-risk Active*. Add a **60-day no-purchase trigger**: churn jumps from 3% to 38% after one missed month, and to 81% after two.
2. **Replace blanket coupons with monthly next-best-action targeting.** Each month, score all members, then send:
3. **service calls** to high-risk members with support issues;
4. **preferred-category coupons** to other high-risk Active members;
5. **win-back offers** to high-value lapsed members.

The expected result is about 84% lower spend, with a profit instead of a loss. 3. **Make service recovery a retention lever.** Members with 2+ tickets churn at 19-37%. Close the loop on tickets within the 2-month fingerprint window, with priority for members the model flags. 4. **Automate free nudges before paid offers.** A 40%+ spend drop or app use below 0.5 sessions/month should trigger app push and personalised content first. 5. **Revisit tier benefits.** Platinum perks are not buying loyalty. Test engagement-based rewards (app streaks, category missions) instead of spend-only tiers. 6. **Measure, don't assume.** Hold out 20% of flagged members as a control group, which needs about 729 flagged members to detect a 15% uplift (section 9.5). Measure the true uplift of each action and feed it back into `config.py`. 7. **Operationalise on the existing stack.** Run a monthly Databricks job (Data Factory ingest → Delta → `spark_features.py` → MLflow champion model → CRM queue). Put the Retention Console on Databricks Apps. Monitor PR-AUC, calibration and PSI drift each month, and shadow-score the GRU challenger.

11. Assumptions

1. **Target.** CHURNED = no transactions in Apr-Jun 2024 for a member who purchased before April (official label, used as-is). Members who never purchased (120 in the label table), the 6 profiled members with no activity, and 5 labelled churners with no pre-April activity cannot be scored from history and are excluded, leaving 2,368 eligible members.
2. **Missing months.** A month with no row *between* a member's first and last rows is lost data, not inactivity (the label table's MONTHS_OBSERVED counts these months as observed). Months *after* a member's last row are treated as zero activity, which is what a live system would see at scoring time.
3. **Spend repair.** $TOTAL_SPEND = TRANSACTIONS \times AVG_BASKET_VALUE$ holds for 99.8% of rows, so negative, zero and 9-15x inflated spend values are rebuilt from that identity rather than dropped.
4. **Impossible zero-transaction values.** Categories bought and coupons redeemed in months with zero transactions are set to 0.
5. **Historical labels.** Snapshots at Jul-23, Oct-23 and Jan-24 reuse the official definition (no purchase in the next 3 months). Applied at Apr-24 this rebuilt label agrees with the official one for 99.8% of members.
6. **Feature window.** All behavioural features use the 6 months before each cut-off, so snapshots are comparable. Recency is capped at 7 months ("6+").
7. **Categoricals.** City and tier variants are standardised by trimming and title-casing (40 to 10 cities, 12 to 3 tiers). Missing gender and price tier become an explicit 'Unknown'. Missing age is filled with the median of the training snapshots, plus a missing flag.
8. **Decision threshold.** 0.37 maximises F1 for Active members of the validation snapshot. High risk means p at or above the threshold; Medium means p of 10% or more.
9. **Economics (not measured, varied in sensitivity analysis).** Value of a retained member = 6 months of their typical monthly spend x 25% gross margin. Coupon: \$10 cost, cuts churn probability by 15% (relative). Service call: \$15, cuts it by 25%. Win-back: \$5, reactivates 5%. Any offer to a Lapsed member is assumed to work only as well as a win-back.
10. **Repeated members.** The same member can appear in up to two training snapshots, so training rows are not fully independent. This is standard for rolling-origin designs, and the test cohort is strictly later in time.

12. Limitations

- **One test quarter.** Results are validated on one forward quarter (with three earlier snapshots for training and validation). Seasonality beyond 18 months is untested.
- **Small sample.** There are only 164 Active churners in the test cohort, so the 95% interval on Active PR-AUC is ± 0.06 . More history would tighten it.
- **Uplifts are assumptions, not measurements.** The scenario values depend on them, which is why break-even points and sensitivity are reported and a controlled test is recommended.
- **SHAP is not causal.** SHAP values and the support what-if describe associations. A member with many tickets may churn because of an underlying problem (a store move, a bad experience) that a phone call cannot fix.
- **Limited data.** There is no transaction-level, product, price, competitor or exit-reason data. Why members leave is inferred from behaviour, not observed.
- **Missing months.** The 953 missing months are treated as lost data. If some were genuine inactivity, a few trend features are slightly optimistic for those members.
- **Correlated features.** Several features are strongly correlated (transactions, spend, active months), so individual logistic coefficients and SHAP shares between correlated features should be read as a group, not one by one.

13. Reproducibility and production path

- `python run_pipeline.py` regenerates every output, chart, the executed notebook, this report and the slide deck (about 1 minute).
- Fixed seeds, pinned `requirements.txt`, and MLflow tracking for every model run (`outputs/mlflow_runs_summary.csv`).
- `python -m pytest`: unit tests, including an **automated leakage test** that corrupts all post-cut-off data and asserts the features do not change.
- **Databricks / Spark:** `src/spark_features.py` is a PySpark port of the feature builder. Verified: identical outputs for 8,302 member-snapshots across 4 cut-offs (31 feature columns, zero mismatches). `notebooks/02_databricks_churn_job.py` shows the scheduled job on Azure Data Factory, Delta tables and the MLflow registry.
- **API:** `uvicorn src.api:app` serves `/score` for real-time scoring, for example from the CRM.
- **Dashboard:** `python app.py` (Dash). **CI:** GitHub Actions rebuilds the outputs and runs the tests on every push. **Governance:** `MODEL_CARD.md` covers intended use, metrics, fairness and monitoring.